

BI:TASK-02:Exploratory Data Analysis (EDA) using Airbnb Dataset

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The errors in the name column are removed and the dataset is loaded

id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price	minimum_nights
1965712	Lovely large 1-bdrm apt UWS	10153235	Tammie	Manhattan	Upper West Side	40.78122	-73.98473	Entire home/apt	235	1
1979579	Best Central Park/NYC View Apt	8219931	Jason	Manhattan	Upper West Side	40.78468	-73.97137	Entire home/apt	300	1
2003807	UWS MANHATTAN APT FOR SUPERBOWL WE	10317189	Edouard	Manhattan	Upper West Side	40.79321	-73.9694	Entire home/apt	1000	1
2103480	Thanksgiving in Manhattan!	10725397	Andrea	Manhattan	Upper West Side	40.77822	-73.98016	Entire home/apt	560	1
2107372	Best NYC Deal on upper west!	10346860	Shlom And Lyndz	Manhattan	Upper West Side	40.79994	-73.97094	Entire home/apt	400	1
2135766	Super Bowl, 2 bdrm, UWS apartment	4074918	Julie	Manhattan	Upper West Side	40.79478	-73.97474	Entire home/apt	900	1
2219294	Great Studio!	10485182	Jennifer	Manhattan	Upper West Side	40.78835	-73.97257	Entire home/apt	60	1
224896	NYC SuperBowl Wk 5 Bdrs River View	11353904	Todd	Manhattan	Upper West Side	40.79476	-73.97299	Entire home/apt	4000	1
2243769	Super Bowl New York City Apartment	11460768	Brian	Manhattan	Upper West Side	40.8002	-73.96045	Entire home/apt	1500	1
2249878	Great UWS Apt./Central Park	470870	Ruben	Manhattan	Upper West Side	40.78531	-73.96988	Entire home/apt	375	1
2249928	Super Bowl Wknd! 3-Bedroom Apt UWS	2675913	Sam	Manhattan	Upper West Side	40.78574	-73.9764	Entire home/apt	1000	1
2273019	Super Bowl Weekend	8105524	Rhett	Manhattan	Upper West Side	40.77018	-73.98643	Entire home/apt	150	1
2283555	SUPER BOWL RENTAL NYC LOW PRICE	11674823	Bruce	Manhattan	Upper West Side	40.77859	-73.97964	Entire home/apt	750	1
2329047	Large 2 Bedroom available	9754117	David	Manhattan	Upper West Side	40.77794	-73.97699	Entire home/apt	400	1
3265275	2BR, UWS, Doorman, Balcony, CntrlPK	16512899	Jeremy	Manhattan	Upper West Side	40.7898	-73.9708	Entire home/apt	200	1
3703037	SUBLET in the UWS	18876264	Daniel	Manhattan	Upper West Side	40.77817	-73.97844	Entire home/apt	200	1
4163804	Classic UWS 1 Bed in Doorman Bldg	15573381	Lindsay	Manhattan	Upper West Side	40.78085	-73.98471	Entire home/apt	250	1
4233705	looking for someone to stay 12days	11243583	Joseph	Manhattan	Upper West Side	40.77715	-73.98235	Entire home/apt	260	1
4262120	Columbus Circle and Park Views	6914968	Lisa	Manhattan	Upper West Side	40.7681	-73.98377	Entire home/apt	2695	1
4525475	Manhattan, UWS, Bright, 1BR, & Loft	12149644	Brandon	Manhattan	Upper West Side	40.78304	-73.98146	Entire home/apt	125	1
4620962	Beautiful one bedroom apt 850 sqft!	21685221	Sael	Manhattan	Upper West Side	40.79293	-73.97126	Entire home/apt	199	1
4782252	PENTHOUSE STUDIO WITH LARGE TERRACE	24657274	Daniel	Manhattan	Upper West Side	40.79202	-73.97947	Entire home/apt	175	1
5395116	Studio in UWS- Manhattan	14586567	Paula	Manhattan	Upper West Side	40.79368	-73.97324	Entire home/apt	150	1
5812875	Large studio, view on Central Park	5963594	Geoffroy	Manhattan	Upper West Side	40.77647	-73.98211	Entire home/apt	175	1
6797129	Spacious 1BR & Loft near Park	8924676	Taeler	Manhattan	Upper West Side	40.77824	-73.97701	Entire home/apt	375	1
6869363	Location! Upper West Side	35997540	Luis	Manhattan	Upper West Side	40.78748	-73.98023	Entire home/apt	150	1
6939668	Lovely 1BR on the Upper West Side	36387290	Megan	Manhattan	Upper West Side	40.78147	-73.97806	Entire home/apt	150	1

(a) What is the average price of the listings?

Average of price

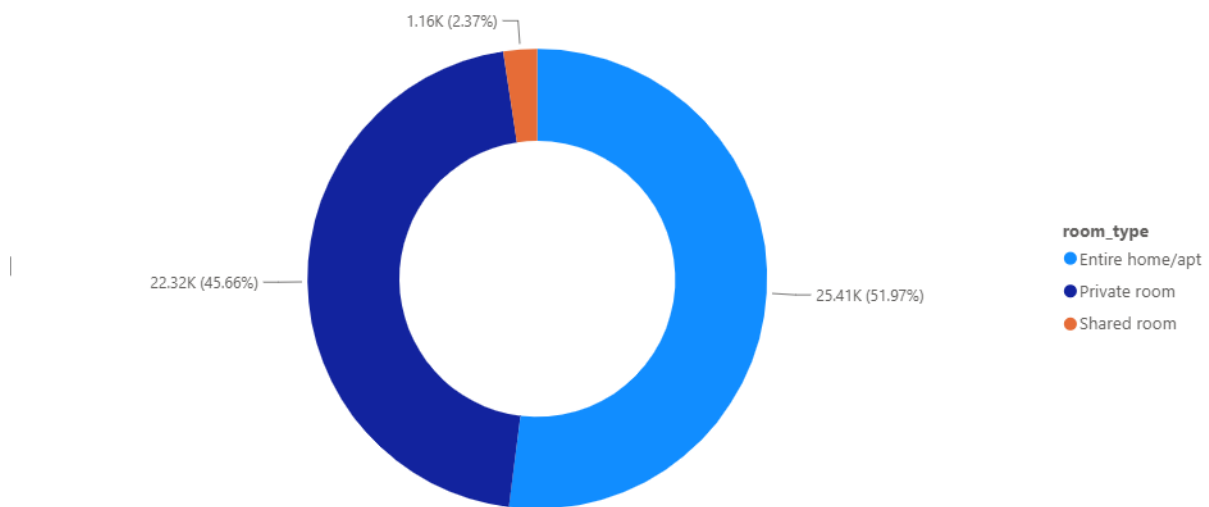
152.72

The average nightly price of Airbnb listings in New York City is \$152.72. This indicates that, on average, guests can expect to pay approximately \$153 per night for an Airbnb stay. Here the table visualization was used.

(b) What is the distribution of the room types available for the listings?

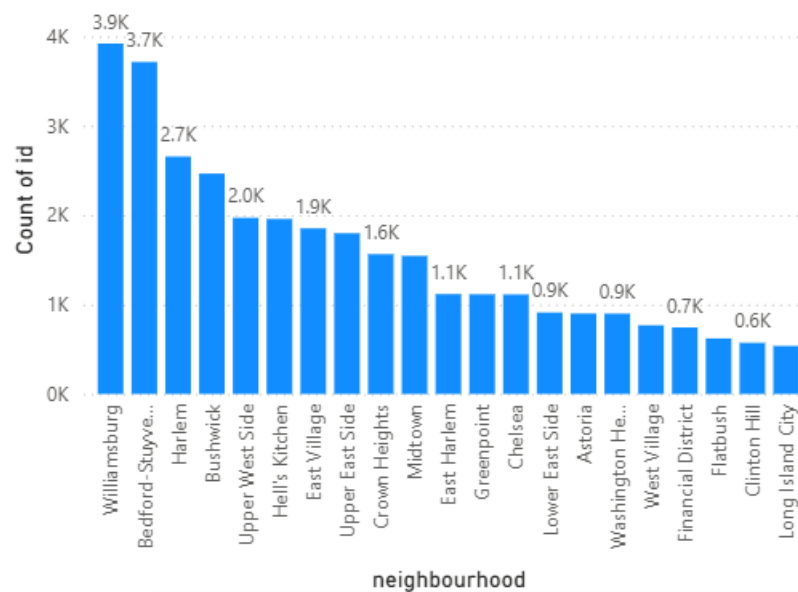
Entire home/apartment listings dominate the Airbnb market in New York City, making up 51.97% of all listings. Private rooms account for 45.66%, while shared rooms represent only 2.37%, making them the least common accommodation type.

Distribution of Airbnb Room Types



(c) Which neighbourhood has the most listings?

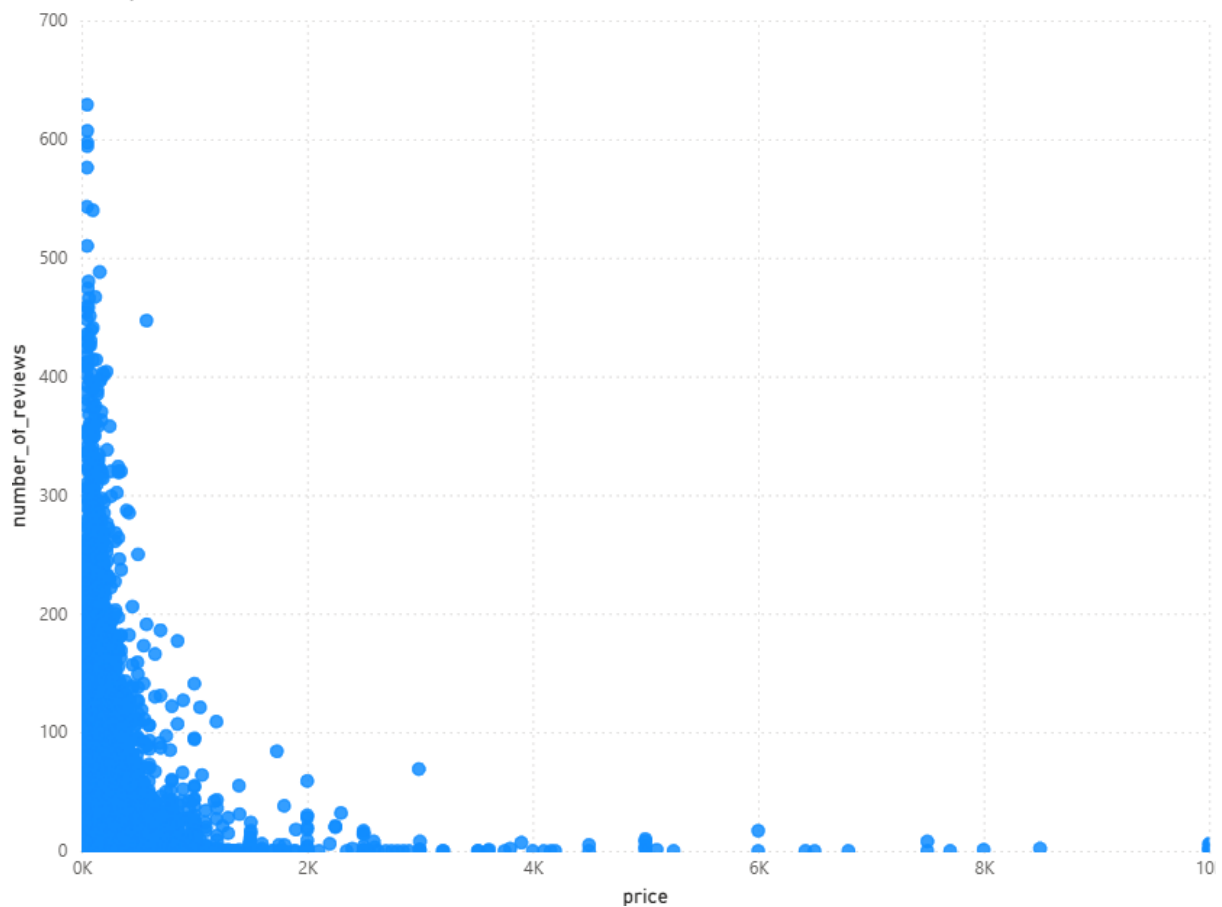
Top 10 Neighbourhoods by Number of Airbnb Listings



Williamsburg has the highest number of Airbnb listings, with approximately 3.9K listings, making it the most popular neighbourhood for Airbnb accommodations in New York City. It is followed by Bedford-Stuyvesant (3.7K) and Harlem (2.7K). The high concentration of listings in these neighbourhoods suggests they are popular locations for both Airbnb hosts and visitors due to factors such as accessibility, tourist attractions, and residential demand.

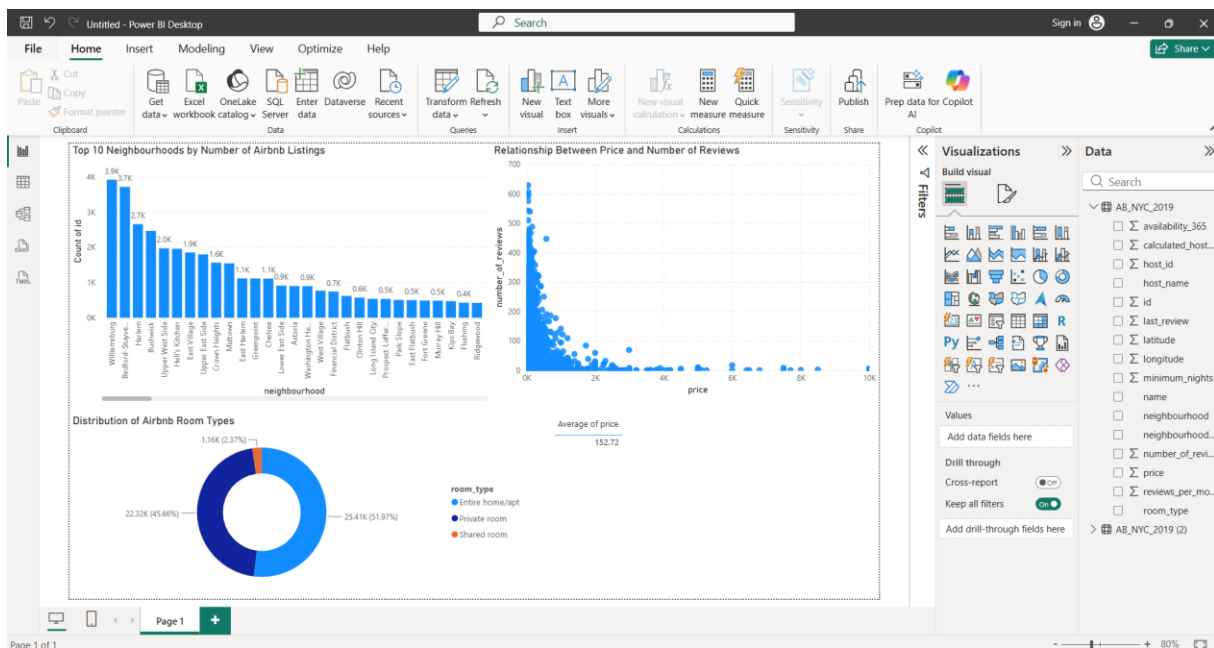
(d) What is the correlation between the number of reviews and the price of the listings?

Relationship Between Price and Number of Reviews



The scatter plot indicates a weak negative correlation between listing price and the number of reviews. Lower-priced Airbnb listings generally receive more reviews, while higher-priced listings tend to receive fewer reviews. This suggests that affordable listings attract more bookings and customer engagement, although the relationship is not strong.

Final Dashboard :



Insights:

The exploratory data analysis of the New York City Airbnb dataset shows that the average listing price is \$152.72 per night. Most listings are Entire homes/apartments, followed closely by Private rooms, while Shared rooms are relatively rare. Williamsburg has the highest concentration of Airbnb listings, indicating its popularity among hosts. Finally, the analysis reveals a weak negative relationship between listing price and the number of reviews, suggesting that more affordable listings generally receive greater customer engagement.